10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 18 — Electromagnetic Induction & Electromagnetic Waves

Solved Exercise | New Book (PECTAA 2026)

Exercise 18 (Solved)

This exercise covers electromagnetic induction, Faraday's law and Lenz's law, the a.c. and d.c. generators, transformers, the electromagnetic spectrum and the particle nature of light.

A. Multiple Choice Questions

18.1 What happens to the induced e.m.f. in a coil if the rate of change of the magnetic field increases?

(a) Decreases
(b) Remains the same
(c) Increases
(d) Becomes zero

Answer: (c) Increases — by Faraday's law the induced e.m.f. is directly proportional to the rate of change of magnetic flux, $E = -N\,\Delta\Phi/\Delta t$, so a faster change gives a larger e.m.f.

18.2 In an A.C generator, slip rings are used to:

(a) reverse current direction periodically
(b) convert A.C to D.C
(c) prevent induced current
(d) increase magnetic flux

Answer: (a) reverse current direction periodically — the two separate slip rings keep each end of the coil connected to the same external terminal, so the current in the external circuit alternates every half rotation.

18.3 The sky appears blue during the day due to:

(a) reflection of light
(b) scattering of shorter wavelengths
(c) absorption of blue light
(d) refraction of light

Answer: (b) scattering of shorter wavelengths — the molecules of the air scatter the short blue wavelengths of sunlight much more strongly than the long red ones (Rayleigh scattering), so blue light reaches our eyes from all directions of the sky.

18.4 Light can be considered as massless particles called:

(a) electrons
(b) protons
(c) photons
(d) neutrons

Answer: (c) photons — light behaves as a stream of massless packets of energy called photons, each carrying energy $E = hf$; this explains the photoelectric effect.

18.5 In a transformer, energy is transferred from the primary coil to the secondary coil through:

(a) direct electrical connection
(b) a magnetic field in the iron core
(c) electrons flowing through the core
(d) heat conduction

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Answer: (b) a magnetic field in the iron core — there is no electrical connection between the coils; the changing magnetic flux set up in the laminated iron core links the two coils and transfers the energy.

B. Short Answer Questions

18.1 What is the direction of induced current in Lenz's law?

Lenz's law states that the direction of the induced current is always such that it opposes the change which produced it. In other words, the induced current sets up a magnetic field that opposes the increase or the decrease of magnetic flux through the coil.

So when a magnet is pushed into a coil (flux increasing), the induced current flows in a direction that makes the face of the coil nearer the magnet a like pole, so the magnet is repelled. When the magnet is pulled out (flux decreasing), the induced current reverses and makes that face an unlike pole, so the magnet is attracted and the loss of flux is resisted. Lenz's law is a direct consequence of the law of conservation of energy.

18.2 Why are slip rings used in an A.C. generator?

The coil of an a.c. generator rotates in a magnetic field, so the current induced in it changes its direction every half rotation. Two separate slip rings, each connected to one end of the coil, rotate with the coil and are in continuous contact with the fixed carbon brushes.

Because each end of the coil is always connected to the same external terminal, the alternating current produced in the coil is delivered to the external circuit as alternating current, changing its direction regularly. If a split-ring commutator were used instead (as in a d.c. generator), the connections would be reversed every half-turn and the output would be direct current. So the slip rings are used to take the a.c. out of the rotating coil without tangling the wires.

18.3 When is induced e.m.f. zero in an A.C generator?

The induced e.m.f. is zero at the instant when the plane of the coil is perpendicular to the magnetic field, that is, when the coil is in the vertical position in the usual diagram. In that position the two sides of the coil move parallel to the field lines, so they do not cut the flux, and the rate of change of flux through the coil is momentarily zero.

By Faraday's law, $E = -N\,\Delta\Phi/\Delta t$, the induced e.m.f. depends on the rate of change of flux and not on the flux itself, so when $\Delta\Phi/\Delta t = 0$ the e.m.f. is zero. This position occurs twice in every rotation and is called the neutral position; the e.m.f. is maximum when the plane of the coil is parallel to the field, because there the sides cut the field lines at right angles.

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18.4 Why is the sky red at sunset?

At sunset the Sun is low on the horizon, so its light has to pass through a much greater thickness of the atmosphere than at noon. On the way, the short wavelengths — blue and green — are scattered away sideways by the molecules and dust of the air (Rayleigh scattering), so they do not reach our eyes directly.

The long wavelengths — red and orange — are scattered much less and therefore travel straight through the thick layer of air to us. That is why the Sun and the sky near it look red or orange at sunrise and sunset. The same reason explains why the sky overhead turns pink and orange while the horizon is red, and why the sky is blue at noon when the light passes through a thinner layer and the scattered blue light reaches us from all over the sky.

18.5 What is a transformer and on what principle does it work?

A transformer is a device that changes (steps up or steps down) the voltage of an alternating current supply. It consists of two coils wound on a common laminated soft-iron core: the primary coil, connected to the input supply, and the secondary coil, connected to the output circuit. The two coils are not electrically connected.

It works on the principle of mutual induction, an application of Faraday's law. When an alternating current flows in the primary coil, the magnetic flux in the iron core keeps changing. This changing flux links the secondary coil and induces an alternating e.m.f. in it. The ratio of the voltages equals the ratio of the numbers of turns: $V_s/V_p = N_s/N_p$.

A transformer works only with alternating current, because only a changing current produces a changing magnetic flux. For an ideal transformer the power stays the same, $V_p I_p = V_s I_s$, so when the voltage is stepped up the current is stepped down.

C. Constructed Response Questions

18.1 If a magnet is moved quickly into a coil, a larger e.m.f. is induced. Why does speed affect the magnitude of the induced current?

By Faraday's law of electromagnetic induction, the induced e.m.f. is proportional to the rate of change of magnetic flux linked with the coil: $E = -N\,\Delta\Phi/\Delta t$. It is not the amount of flux that matters but how fast it changes.

When the magnet is moved quickly, the flux through the coil changes from zero to its full value in a very short time, so $\Delta\Phi/\Delta t$ is large, and the induced e.m.f. is large. Since the resistance of the coil is fixed, the induced current $I = E/R$ is also large. When the same magnet is moved in slowly, the same total change of flux takes a much longer time, $\Delta\Phi/\Delta t$ is small, and both the e.m.f. and the current are small.

This is easy to feel: pushing a magnet quickly into a coil requires more force, because the larger induced current produces a stronger opposing magnetic field (Lenz's law) which resists the motion. If the magnet is held stationary inside the coil, $\Delta\Phi/\Delta t = 0$, so no e.m.f. and no current are produced at all, however strong the field.

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18.2 Why is an iron core used in transformers instead of wood or plastic? Explain briefly.

A transformer transfers energy from the primary to the secondary coil by means of the magnetic flux that links them. The core must therefore carry the magnetic flux from one coil to the other with as little loss as possible.

Soft iron is a ferromagnetic material: it has very high magnetic permeability, so the flux produced by the primary coil is concentrated inside it and almost all of it passes through the secondary coil. It is also easily magnetised and demagnetised, so it follows the alternations of the a.c. without lag and without retaining magnetism. Wood and plastic are non-magnetic (their permeability is nearly that of air), so they cannot carry flux; using them would leave the flux mostly outside the coils and hardly any e.m.f. would be induced, so the transformer would give almost no output.

The core is also made laminated — thin iron sheets separated by an insulating varnish — which reduces energy loss from eddy currents, and the core is made of soft iron rather than steel so that hysteresis loss is small. The coils are wound on top of this core to give tight magnetic coupling.

18.3 How does the concept of photons help explain the working of solar sails in space?

According to the particle nature of light, light is a stream of massless packets of energy called photons. Each photon, though it has no rest mass, carries energy and momentum ($p = E/c = hf/c$). When a photon strikes a surface and is absorbed or reflected, it delivers that momentum to the surface, just as a cricket ball does when it hits a bat. The tiny force so produced is called radiation pressure.

A solar sail is a very large, thin, highly reflecting sheet attached to a spacecraft. Sunlight falling on it consists of an enormous number of photons. Because the sail is reflecting, each photon bounces back and gives up about twice as much momentum as it would if it were simply absorbed. The billions of such impacts each second add up to a small but steady force.

In space there is no air resistance to overcome, so this continuous push slowly accelerates the spacecraft — exactly as wind pushes a sailboat on the sea. In this way a solar sail can move a spacecraft without carrying any fuel. The wave theory, with its continuous energy spread over the wave front, does not explain this pressure in such a simple way, so the photon picture is needed.

18.4 Why can a transformer work only on alternating current and not on direct current?

A transformer works by mutual induction: the changing magnetic flux produced by the primary coil induces an e.m.f. in the secondary coil. Faraday's law shows that the induced e.m.f. depends on the rate of change of flux, $E = -N\,\Delta\Phi/\Delta t$. If there is no change of flux, there is no induced e.m.f. at all.

With direct current the current in the primary coil is steady, so the magnetic field it produces in the core is also constant. The flux linked with the secondary coil does not change, $\Delta\Phi/\Delta t = 0$, and therefore no e.m.f. is induced in the secondary. The transformer gives no output voltage, even though a steady current is flowing in the

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primary.

With alternating current the current, and hence the flux, changes continuously — it rises, falls and reverses 50 times a second. This continuous change induces an alternating e.m.f. in the secondary coil, so the transformer works. (If a steady d.c. were suddenly switched on or off, a momentary pulse would be induced, but a steady d.c. gives no continuous output.)

D. Comprehensive Questions

18.1 Describe how Lenz's Law supports the law of conservation of energy. Give one real-life example to support your explanation.

Lenz's law says that the induced current always flows in such a direction that its magnetic effect opposes the change that produced it. So when we try to push a magnet into a coil, the induced current makes the near face a like pole, and a repulsive force pushes the magnet back; when we pull the magnet out, the induced current reverses and the coil attracts the magnet and tries to hold it. In both cases the induced effect works against what we are doing.

Why this supports conservation of energy: to push the magnet in against this repulsion, we have to do mechanical work, and it is exactly this work that is converted into the electrical energy of the induced current. If the induced current helped the motion instead of opposing it, the magnet would be accelerated and would gain electrical energy as well as kinetic energy from nothing — a perpetual motion machine, which is impossible. So Lenz's law is the statement, in electromagnetic induction, of the fact that energy can never be created; the electrical energy obtained is always paid for by the mechanical work done.

Real-life example: in an induction furnace and an electromagnetic brake, a metal disc spun between the poles of a magnet is brought to rest quickly, because the eddy currents induced in it oppose its motion. The kinetic energy of the disc is converted into heat in the metal, and the disc slows down — energy is conserved. Similarly, the extra force needed to turn the armature of a generator when the current drawn from it is increased shows that the electrical power delivered is taken from the mechanical power supplied.

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18.2 Compare step-up and step-down transformers. Include the difference in number of turns, voltage output, and where each is used.

Point	Step-up transformer	Step-down transformer
Number of turns	$N_s > N_p$ — more turns in the secondary coil	$N_s < N_p$ — fewer turns in the secondary coil
Output voltage	$V_s > V_p$, the voltage is increased	$V_s < V_p$, the voltage is decreased
Output current	smaller than the input current ($I_s < I_p$)	larger than the input current ($I_s > I_p$)
Formula	$V_s/V_p = N_s/N_p$, with $N_s/N_p > 1$	$V_s/V_p = N_s/N_p$, with $N_s/N_p < 1$

Point	Step-up transformer	Step-down transformer
Wire of coils	primary of thick wire, secondary of thin wire (fewer turns)	primary of thin wire, secondary of thick wire
Where used	at the power station, to raise the voltage for transmission, and in the high-voltage part of a TV or X-ray set	near homes and factories, to bring the voltage down to 220 V, and in chargers, adaptors and doorbells

Example: 500 primary turns and 100 secondary turns with 220 V input gives V_s = 220 × 100/500 = 44 V — a step-down transformer. A turns ratio of 2 : 100 with 25 V input gives 1250 V — a step-up transformer. In both types the power is nearly conserved, $V_p I_p = V_s I_s$, so when the voltage goes up the current comes down and vice versa.

18.3 How does the scattering of sunlight cause the sky to appear blue during the day and red at sunset? Explain.

Sunlight is white light containing all the colours, from violet andblue of short wavelength (about 400 nm) to red of long wavelength (about 700 nm). When it enters the atmosphere it meets molecules of nitrogen and oxygen and tiny dust particles whose size is much smaller than the wavelength of light. Such particles scatter light of shorter wavelength much more strongly than light of longer wavelength — the scattered intensity goes as $1/\lambda^4$ (Rayleigh scattering).

Blue during the day. Of all the colours in sunlight, blue is scattered most. So the blue light is scattered in all directions by the air molecules and reaches our eyes from every part of the sky, while the red light passes on almost straight. This is why the sky looks blue when we look away from the Sun. (Violet is scattered even more, but our eyes are less sensitive to it and part of it is absorbed in the upper atmosphere, so the sky appears blue rather than violet.)

Red at sunset. When the Sun is near the horizon, its rays travel a much longer path through the atmosphere. Almost all the blue and green light is scattered out of the beam on the way, so the light that finally reaches us directly is mainly of long wavelength — red and orange. That is why the Sun looks red and the sky around it is red or orange at sunrise and sunset. The same effect makes distant mountains look bluish and the sky red after a volcanic eruption, when there is more dust in the air.

18.4 What is an oscilloscope, and how does it help in analyzing electrical signals? Describe its main components.

An oscilloscope (cathode-ray oscilloscope, CRO) is an instrument that displays a graph of voltage against time on a fluorescent screen, so that the shape, amplitude and frequency of an electrical signal can be seen and measured directly.

Main components: (i) the cathode-ray tube — an evacuated glass tube with an electron gun (a heated cathode) that produces and accelerates a beam of electrons; (ii) the focusing and accelerating anodes, which make the beam narrow and fast; (iii) two pairs of deflecting plates — the Y-plates deflect the beam vertically in proportion to the input signal voltage and the X-plates sweep it horizontally at a steady rate to

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give the time base; (iv) the fluorescent screen coated with zinc sulphide, which glows where the beam strikes it; and (v) the control circuits — the time-base generator, amplifiers for the X and Y inputs, the trigger circuit and the power supply. How it helps. The spot moves across the screen at a known, uniform rate, so the horizontal axis becomes a time axis. A signal applied to the Y-plates moves the spot up and down, so a waveform is drawn. From the screen we read: the amplitude (peak voltage) using the volts-per-division setting; the time period and hence the frequency using the time-per-division setting; the shape of the wave, showing distortion, noise and phase difference; and with two traces, the gain and phase of an amplifier. In this way an oscilloscope is used to test radio and TV circuits, heart monitors, sound equipment and any pulse or waveform in the laboratory.

18.5 What is the shape of the path followed by an electron when it moves at right angles to a uniform magnetic field? Explain why.

The path is a circle, in a plane perpendicular to the magnetic field, and the electron moves with a constant speed round it.

Reason: when a charged particle moves in a magnetic field, it experiences a force F = q v B sin θ. As the velocity is at right angles to the field, θ = 90° and the force is F = q v B, which is always perpendicular both to the velocity and to the field — so it acts as a centripetal force. A force of constant magnitude acting always at right angles to the velocity changes only the direction of motion, not its speed, and the resulting path is a circle.

The radius of the circle follows from centripetal force = magnetic force, $m v^2/r = q v B$, so $r = m v / (q B)$. A faster electron or a weaker field gives a bigger circle; a stronger field or a larger charge gives a smaller one. The direction of rotation is given by Fleming's left-hand rule (reversed for a negative charge). This principle is used in the cyclotron, in mass spectrometers and in magnetic focusing in television tubes and electron microscopes.

E. Numerical Problems

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18.1 A coil consisting of 50 tightly wound turns experiences a change in magnetic flux from 0.3 Wb to 0.8 Wb over a time interval of 0.4 s. Determine the average induced electromotive force (e.m.f.) generated in the coil during this time.

Given: N = 50 turns
$\Phi_1 = 0.3$ Wb, $\Phi_2 = 0.8$ Wb
$\Delta\Phi = 0.8 - 0.3 = 0.5$ Wb
$\Delta t = 0.4$ s

To find: E = ?

Formula: $E = -N(\Delta\Phi / \Delta t)$

Solution:
$E = -50 \times (0.5 / 0.4)$
$E = -50 \times 1.25$
$E = -62.5$ V

Answer: E = −62.5 V (the minus sign shows the e.m.f. opposes the change, Lenz's law)

18.2 A circular loop of radius 0.2 m and resistance 5 Ω is placed perpendicular to a magnetic field of 0.5 T. If the field drops to zero in 0.1 s, calculate: (a) the induced emf (b) the induced current and its direction (assuming the field points into the page).

Given: r = 0.2 m
$A = \pi r^2 = 3.14 \times (0.2)^2 = 0.1256$ m^2
B = 0.5 T → 0, ΔB = 0.5 T
Δt = 0.1 s
R = 5 Ω

To find: (a) E = ? (b) I = ?

Formula: $\Phi = B A$, $E = -\Delta\Phi/\Delta t = -A\,\Delta B/\Delta t$, $I = E / R$

Solution:
$\Delta\Phi = A\,\Delta B = 0.1256 \times 0.5 = 0.0628$ Wb
(a) $E = 0.0628 / 0.1 = 0.628$ V
$E \approx 0.63$ V
(b) $I = 0.63 / 5 = 0.126$ A
The field into the page is decreasing, so the induced current produces a field into the page — the current is clockwise.

Answer: (a) E = 0.63 V (b) I = 0.126 A, clockwise

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18.3 A transformer has 500 primary turns and 100 secondary turns. If the primary voltage is 220 V, find: (a) secondary voltage (b) is this a step-up or step-down transformer?

Given: $N_p = 500$
$N_s = 100$
$V_p = 220$ V

To find: (a) V_s = ? (b) type = ?

Formula: $V_s / V_p = N_s / N_p$

Solution:
$V_s = V_p \times (N_s / N_p)$
$V_s = 220 \times (100 / 500)$
$V_s = 220 \times 0.2$
$V_s = 44$ V
$N_s < N_p$, so the voltage is reduced.

Answer: (a) $V_s = 44$ V (b) step-down transformer ($V_s < V_p$)

18.4 A step-up transformer has a turns ratio of 2:100. If an alternating voltage of 25 V is applied to the primary coil, what will be the secondary voltage?

Given: $N_p : N_s = 2 : 100$
$V_p = 25$ V

To find: V_s = ?

Formula: $V_s / V_p = N_s / N_p$

Solution:
$V_s = V_p \times (N_s / N_p)$
$V_s = 25 \times (100 / 2)$
$V_s = 25 \times 50$
$V_s = 1250$ V

Answer: $V_s = 1250$ V

18.5 A step-down transformer has a turns ratio of 100:5. If an A.C voltage of 190 V is applied to the primary coil, what is the voltage across the secondary coil?

Given: $N_p : N_s = 100 : 5$
$V_p = 190$ V

To find: V_s = ?

Formula: $V_s / V_p = N_s / N_p$

Solution:
$V_s = V_p \times (N_s / N_p)$
$V_s = 190 \times (5 / 100)$
$V_s = 190 \times 0.05$
$V_s = 9.5$ V

Answer: $V_s = 9.5$ V

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Formulae to Remember

- Faraday's law: $E = -N\,\Delta\Phi/\Delta t$ ($\Phi = B\,A$)
- Lenz's law: the induced current opposes the change that produced it.
- Transformer: $V_s/V_p = N_s/N_p = I_p/I_s$
- Photon energy: $E = hf$; momentum $p = hf/c$
- Electron in a magnetic field: $r = m\,v\,/\,(q\,B)$

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